

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2023
CHE-1104: Chemistry

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Define electron affinity. Discuss the trends of electron affinity across a period and down a group of the periodic table. 4
 b) What are quantum numbers? Explain the physical significance of the four quantum numbers. 4
 c) Write down the detail explanation that second ionization potential of lithium is higher than its first ionization potential. 4
 d) Explain Heisenberg uncertainty principle. Show that the principle is valid only for small particles and not for large particles. 2

2. a) What is polar covalent bond? Explain with example. 3
 b) Why coordinate bond is a special type of covalent bond? 3
 c) Explain sp^2 hybridization for $CH_2=CH_2$ with diagram. 4
 d) What is VSEPR theory? Predict the shape of CH_4 , NH_3 and XeF_6 based on VSEPR theory. 4

3. a) State and explain Raoult's law. 4
 b) Describe a method for determining the molar mass of a non-volatile solute by the relative lowering in vapour pressure of the solvent. 6
 c) How does sprinkling of salt help in clearing the snow covered roads in hilly areas? Explain the phenomenon involved in the process. 4

4. a) What is K_p and K_c ? How K_p and K_c are related? 4
 b) Explain how equilibrium constant changes with temperature for exothermic and endothermic reaction. 4
 c) What is Le Chatelier Principle? The reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + \text{heat}$ is at equilibrium. State and explain the effects of
 (i) Addition of H_2 in the system
 (ii) Increase in pressure and
 (iii) Decrease in temperature on the position of equilibrium. 6

5. a) What do you understand by ionic product of water? How p^H scale can be established from the ionic product of water? 4
 b) Explain conjugate acid and conjugate base with example. 2
 c) Explain with an example why pH of a buffer solution does not change significantly on small addition of acids or bases. 4
 d) What is buffer action? Calculate the p^H value of 0.1(M) CH_3COOH solution ($K_a = 1.8 \times 10^{-5}$). 4

6. a) Deduce the rate equation for the type of following reaction: $2A \Rightarrow \text{products}$. 4
 b) Define order and molecularity of a reaction. 4
 c) It needs 30 minutes for 25% completion of a 1st order reaction.
 Find: i) Rate constant ii) Half - life. 3
 d) Show that 1st order reaction never completes. 3

7. a) With an appropriate diagram, define standard electrode potential and state the differences between Electrochemical cell and Electrolytic cell. 5
 b) Discuss the enthalpy of a reaction and state Hess's law of thermochemistry. 5
 c) What is Gibb's phase rule? With a phase diagram, enlighten the mono component system like H_2O . 4